

ABDUL RAZAK

Staff Data Scientist, Extreme Networks

Salem, United States | Portfolio | Google Scholar | DBLP | ORCID | Patents | LinkedIn

RESEARCH

My work concerns decision making when objectives conflict and preferences are known only in part. During my doctorate and post-doctoral fellowship I developed exact and approximate algorithms for multi-objective constraint optimisation and influence diagrams, and methods that elicit only as much preference information as a decision actually requires; this appeared at AAAI, IJCAI, UAI, CP and ICTAI, largely with colleagues at IBM Research Dublin. In industry I have applied the same probabilistic and decision-theoretic tools to building security, where the work produced eight granted patents, and since 2020 to enterprise wireless networks, where I lead the data-science work on generative models for network diagnosis.

INTERESTS Multi-objective decision making under uncertainty; preference elicitation and inference; probabilistic graphical models and influence diagrams; grounded and retrieval-augmented generation; constraint optimisation; machine learning for networked systems.

EDUCATION

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| 2010–2014 | PhD, Computer Science (Artificial Intelligence)
National University of Ireland, Cork — Cork Constraint Computation Centre
Thesis: <i>Reasoning with Imprecise Trade-offs in Decision Making under Certainty and Uncertainty</i> . Supervised by Nic Wilson. Funded by the IRCSET / IBM Enterprise Partnership Scheme as part of IBM's Risk Management Colloboratory. |
| 2003–2005 | MSc, Mathematics with Computer Science
Osmania University, Hyderabad, India
Awarded the university merit scholarship. |
| 2000–2003 | BSc, Mathematics, Statistics and Computer Science
Osmania University, Hyderabad, India |

APPOINTMENTS

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| 2020–present | Staff Data Scientist
Extreme Networks — Salem, United States; previously Cork, Ireland <ul style="list-style-type: none">Lead the data-science work on generative models for network diagnosis. Language models are used for hypothesis generation and explanation; anomaly detection, a knowledge graph of the deployment and live telemetry supply the evidence they reason over.Built agent workflows that triage client connectivity, roaming and device-health problems across deployments of thousands of access points and switches, and draft remediation steps for network operators.Developed correlation and inference pipelines over telemetry and client-experience metrics, so that investigation of routine faults takes seconds rather than hours.Adapted the preference-inference methods from my research to rank recommendations for customers whose priorities differ in reliability, cost and coverage. |
| 2017–2020 | Data Scientist
Johnson Controls — Cork, Ireland <ul style="list-style-type: none">Machine-learning lead for the intrusion-alarm, access-control and retail-analytics product lines.Alarm-panel classification by Pareto optimisation, root-cause analysis of events, and site-independent signature generation for predictive security analysis; eight granted US patents issued from this work.Relational probabilistic models in BLOG and Bayesian inference in PyMC3, deployed across large multi-site estates. |
| 2014–2017 | Post-Doctoral Fellow (Science Foundation Ireland)
Insight Centre for Data Analytics, NUI Cork <ul style="list-style-type: none">Probabilistic graphical models, clustering and influence diagrams for industrial decision problems, within a programme run with United Technology Research Centre.Preference inference with lexicographic and weighted-average models, partly with IBM Research Dublin (AAAI 2017, IJCAI 2015, ICTAI 2015). |
| 2005–2010 | Lecturer in Mathematics
Osmania University — Hyderabad, India <ul style="list-style-type: none">Undergraduate teaching: abstract and linear algebra, real and numerical analysis, differential and vector calculus, probability and statistics. |

PUBLICATIONS

Peer-reviewed; complete list on Google Scholar and DBLP.

- [C6] R. Marinescu, **A. Razak**, N. Wilson. Multi-objective influence diagrams with possibly optimal policies. *Proc. 31st AAAI Conference on Artificial Intelligence*, 2017.
- [C5] Anne-Marie, **A. Razak**, N. Wilson. The comparison of multi-objective preference inference based on lexicographic and weighted average models. *Proc. 27th IEEE International Conference on Tools with Artificial Intelligence (ICTAI)*, 2015.
- [C4] N. Wilson, **A. Razak**, R. Marinescu. Computing possibly optimal solutions for multi-objective constraint optimisation with trade-offs. *Proc. 24th International Joint Conference on Artificial Intelligence (IJCAI)*, 2015.
- [W1] Anne-Marie, **A. Razak**, N. Wilson. Multi-objective constraint optimisation with lexicographic preference models. *Workshop on Algorithmic Issues for Inference in Graphical Models (AIGM)*, 2015.
- [C3] R. Marinescu, **A. Razak**, N. Wilson. Multi-objective constraint optimisation with trade-offs. *Proc. 19th International Conference on Principles and Practice of Constraint Programming (CP)*, 2013.
- [C2] R. Marinescu, **A. Razak**, N. Wilson. Multi-objective influence diagrams with trade-offs. *Proc. 28th Conference on Uncertainty in Artificial Intelligence (UAI)*, 2012.

GRANTED US PATENTS

Nine granted between 2019 and 2021; filings searchable on Justia and Google Patents.

DECISION INTELLIGENCE AND MAINTENANCE

- [P9] **A. Razak**, L. A. Deleris, R. Marinescu, N. Wilson. Interactive method to reduce the amount of trade-off information required from decision-maker in multi-attribute decision-making under uncertainty. *Granted 2019*.
- [P8] **A. Razak**, G. Subramanian, C. J. Donovan. Virtual maintenance manager. *Granted 2019*.
- [P7] D. Horgan, **A. Razak**, G. Bernardi. Root-cause analysis of event occurrences. *Granted 2021*.

ALARM MANAGEMENT

- [P6] **A. Razak**, M. Stewart, G. Subramanian. Automated alarm panel classification using Pareto optimization. *Granted 2020*.
- [P5] **A. Razak**. Systems and methods for managing alarm data of multiple locations. *Granted 2021*.
- [P4] M. Costello, F. Fala, J. R. Holliday, N. B. Pavel, **A. Razak**. Building security system with alarm lists and alarm rule editing. *Granted 2021*.

BUILDING SECURITY ANALYTICS

- [P3] M. Dziduch, C. J. Donovan, **A. Razak**. Building security analysis system with site-independent signature generation for predictive security analysis. *Granted 2020*.
- [P2] M. Dziduch, C. J. Donovan, **A. Razak**. Rules-based method of identifying misuse of emergency fire exits using data generated by a security alarm system. *Granted 2020*.
- [P1] M. Dziduch, C. J. Donovan, **A. Razak**. Systems and methods for managing a security system. *Granted 2020*.

HONOURS AND FUNDING

- IEEE Senior Member, elevated May 2026.
- Science Foundation Ireland post-doctoral fellowship, Insight Centre for Data Analytics, NUI Cork, 2014–2017.
- IRCSET / IBM Enterprise Partnership Scheme scholarship; full doctoral funding, 2010–2013.
- Osmania University merit scholarship (MSc, 2003–2005); ranked 40th of roughly 30,000 candidates in the university mathematics entrance examination.

SERVICE AND TALKS

- Reviewer, *Computers and Electrical Engineering* (Elsevier); more than ten manuscripts in artificial intelligence, wireless networking and intelligent systems.
- Reviewer, IEEE International Conference on Communications (ICC); artificial-intelligence submissions to the 2025 technical programme.
- Invited talks: Sapienza University of Rome, on preference handling in decision making under certainty and uncertainty; NUI Galway, on probabilistic decision making under uncertainty; NUI Cork and IBM Research Dublin, during the Risk Management Collobratory and United Technology Research Centre programmes.
- Conference presentations, oral and poster, at AAAI, IJCAI, UAI, CP and ICTAI.
- Five years of undergraduate lecturing; mentoring of junior data scientists in industry.

TOOLS

PROGRAMMING	Python, C++ (STL, Boost, Eigen), SQL, PySpark, PyFlink.
MODELLING	PyMC3, BLOG, Swift, ILOG CPLEX, LPSOLVE, scikit-learn.
PLATFORMS	AWS SageMaker and Athena, Azure Databricks, streaming and batch pipelines.
LANGUAGES	English, Hindi, Telugu, Urdu. Irish citizen since 2021.